Do Now · retrieval

Lesson 118 · Extreme Programming and RAD

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which lifecycle uses relatively fixed sequential stages?

Q2

For the binary value 1010, what is its decimal value?

Q3

A linked-list node has data and what other essential field?

Q4

In a BST, where is a value less than the current node placed?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

XP practices — Test-driven development, pair programming, refactoring and continuous integration.

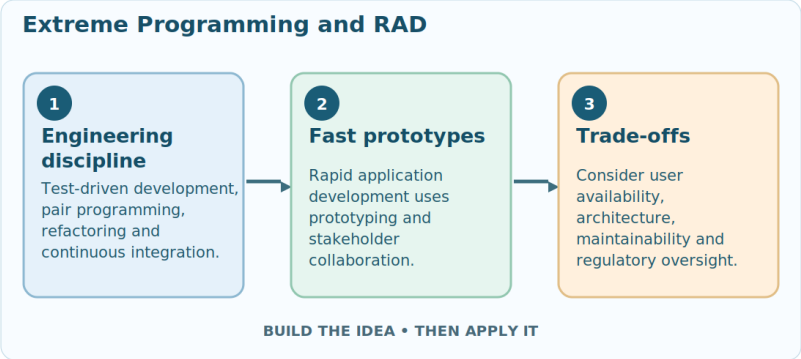
Core idea 2

RAD approach — Rapid application development uses prototyping and stakeholder collaboration.

Core idea 3

Balance cost and fit — Consider user availability, architecture, maintainability and regulatory oversight.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

XP — Agile method emphasising disciplined engineering practices.

TDD — Write a failing test then implement/refactor.

RAD — Development centred on rapid prototyping and user feedback.

refactoring — Improving structure without changing observable behaviour.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

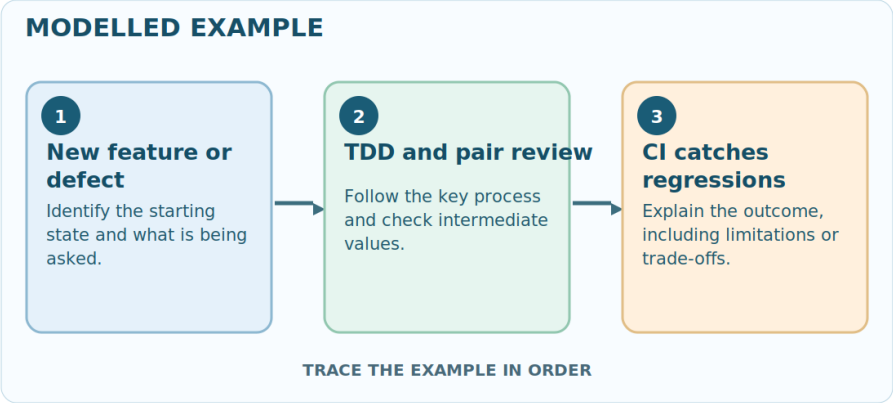
Problem

A small team repeatedly breaks existing features after changes. Which XP practices help?

Follow the worked method

1. Model answer — Automated tests written before/with code (TDD) define intended behaviour, continuous integration runs regression tests on each change, and refactoring improves design without changing observable behaviour. — Check: link each claim to the scenario and use precise OCR terminology.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain test-driven development and pair programming within XP. [6]

Question 2

Describe two reasons RAD may suit a system with unclear interface requirements. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate XP and RAD for building a safety-critical medication scheduler under time pressure. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Refactoring is intentionally changing the program's externally visible requirements.

Use key vocabulary from memory

1. Explain XP in your own words.

2. Explain TDD in your own words.

3. Explain RAD in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.